2.1.6 Polysaccharides (Non Sugars)

Polysaccharides are the carbohydrates that yield more than ten monosaccharide units upon hydrolysis. They are further classified into homopoly saccharides and heteropoly saccharides based on the monomeric units. Example: starch, cellulose, inulin

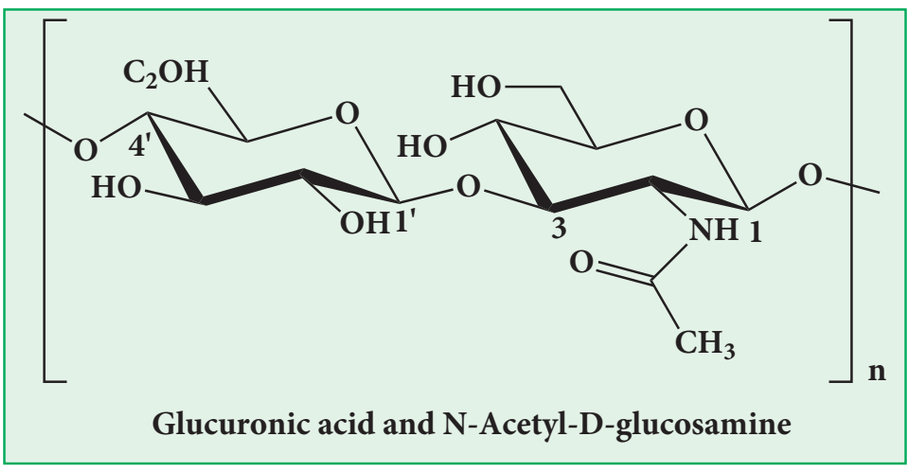
Homopolysaccharides

A homopolysaccharide yields the same type of monosaccharide units on hydrolysis. For example starch a homopolysaccharide yields only glucose upon hydrolysis. Similarly glycogen and cellulose also yield glucose on hydrolysis.

Heteropolysaccharides

A heteropolysaccharide yields more than one type of monosaccharide upon hydrolysis. Eg. Hyaluronic acid, heparin, keratan sulphate and chondroitin sulphate

These are present in extra cellular matrix and therefore they are called as mucopolysaccharides.



Glucuronic acid and N-Acetyl-D-glucosamine

Figure 2.5 Structure of hyaluronic acid

Hyaluronic acid is made up of glucuronic acid and N-acetyl glucosamine.

Starch:

Starch is the major form of stored carbohydrate in plants. Starch is composed of a mixture of two substances namely amylose, a linear polysaccharide and amylopectin, a branched polysaccharide. The detailed study of starch will be discussed in the unit 5.

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